

HR Employee Attrition Analysis & Interactive Dashboard

1. Executive Summary

Employee attrition is one of the major challenges faced by organizations because high turnover increases recruitment costs, decreases productivity, and affects overall employee morale. This project focuses on analyzing employee attrition patterns using HR analytics techniques to identify the major factors influencing employee turnover.

The project involved data cleaning, preprocessing, exploratory data analysis, SQL-based business analysis, Excel analytics, and the development of an interactive Power BI dashboard. Various business metrics such as attrition rate, salary trends, overtime impact, and department-wise turnover were analyzed to generate actionable business insights.

The analysis revealed that overtime, lower salary ranges, limited career growth, and specific departments contribute significantly to employee attrition. The dashboard provides HR managers with an interactive tool to monitor workforce trends and make data-driven decisions for improving employee retention.

2. Introduction

Human Resource Analytics plays an important role in helping organizations understand workforce behavior and employee retention trends. Employee attrition can lead to increased operational costs, reduced efficiency, and loss of experienced talent. Therefore, understanding the factors responsible for attrition is essential for improving workforce stability.

This project aims to analyze HR employee data and uncover meaningful insights related to employee turnover. By combining Python, SQL, Excel, and Power BI, the project demonstrates a complete end-to-end analytics workflow used in real business environments.

3. Business Problem

Organizations often struggle to identify why employees leave the company. High employee turnover affects:

- Productivity
- Recruitment costs
- Employee morale
- Business continuity

The HR department requires a data-driven solution to:

- Identify high-risk employees
- Understand attrition patterns
- Improve retention strategies
- Support workforce planning

4. Project Objectives

The primary objectives of this project are:

- Analyze employee attrition trends
- Identify factors affecting employee turnover
- Perform HR business analysis using SQL
- Create interactive dashboards for decision-making
- Generate business recommendations to improve employee retention

5. Dataset Information

The dataset used for this project contains HR employee records including demographic, professional, and performance-related information.

Dataset Features Include:

- Age
- Gender
- Department
- Job Role
- Monthly Income
- Overtime
- Work-Life Balance
- Years at Company
- Job Satisfaction
- Attrition Status

Target Variable:

- Attrition (Yes:1/No:0)

The dataset was cleaned and transformed before performing analysis.

6. Tools & Technologies

Tool	Purpose
Python	Data Cleaning & EDA
Pandas & NumPy	Data Manipulation
Matplotlib & Seaborn	Data Visualization
MySQL	Business Analysis
Excel	KPI Reporting

Tool	Purpose
Power BI	Interactive Dashboard
GitHub	Project Hosting

7. Data Cleaning & Preprocessing

Several preprocessing techniques were applied to improve data quality and ensure accurate analysis.

Cleaning Steps:

- Removed duplicate records
- Handled missing values
- Corrected data types
- Encoded categorical variables
- Standardized column names
- Created additional business metrics

Feature Engineering:

- Age Groups
- Salary Bands
- Attrition Indicators
- Employee Risk Categories
- Department
- EducationField
- Gender
- Attrition Percentage
- Employee Tenure Group
- BusinessTravel
- Experience Level
- JobRole

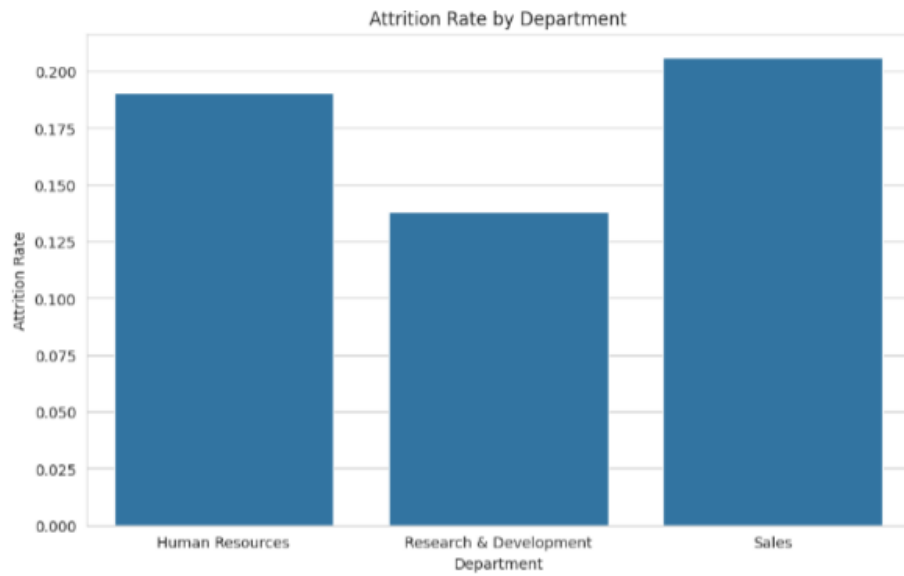
These preprocessing steps improved dataset consistency and analysis accuracy.

8. Exploratory Data Analysis (EDA)

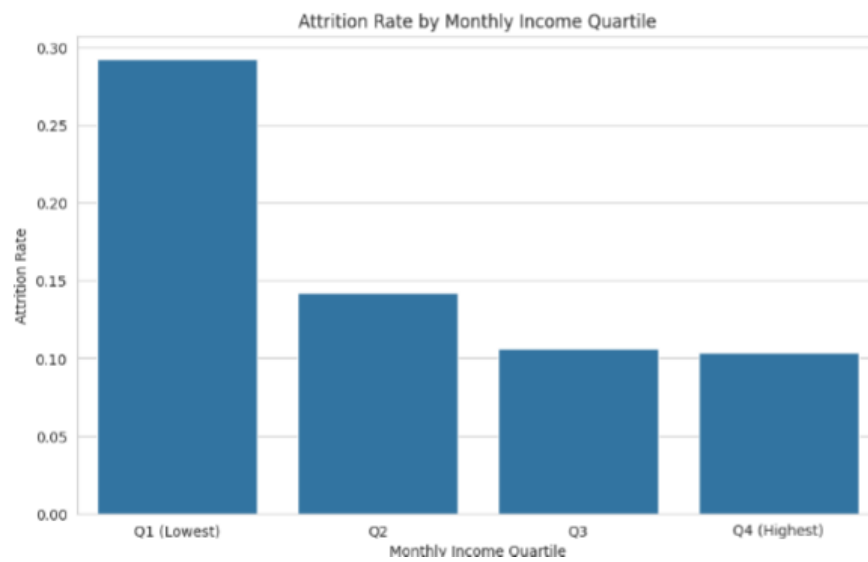
EDA was performed to understand employee behavior patterns and identify important trends.

Key Analysis Performed:

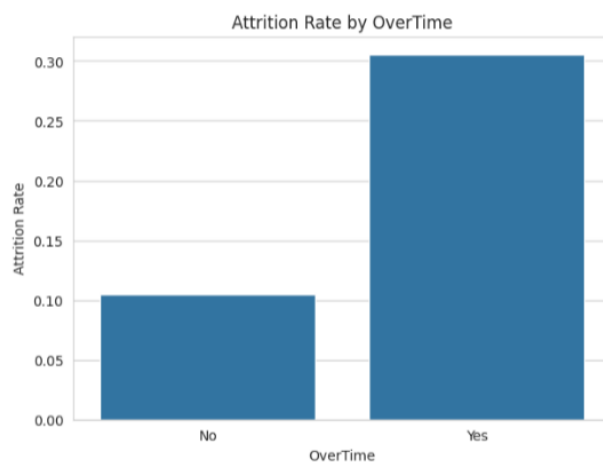
- Attrition Rate by Department



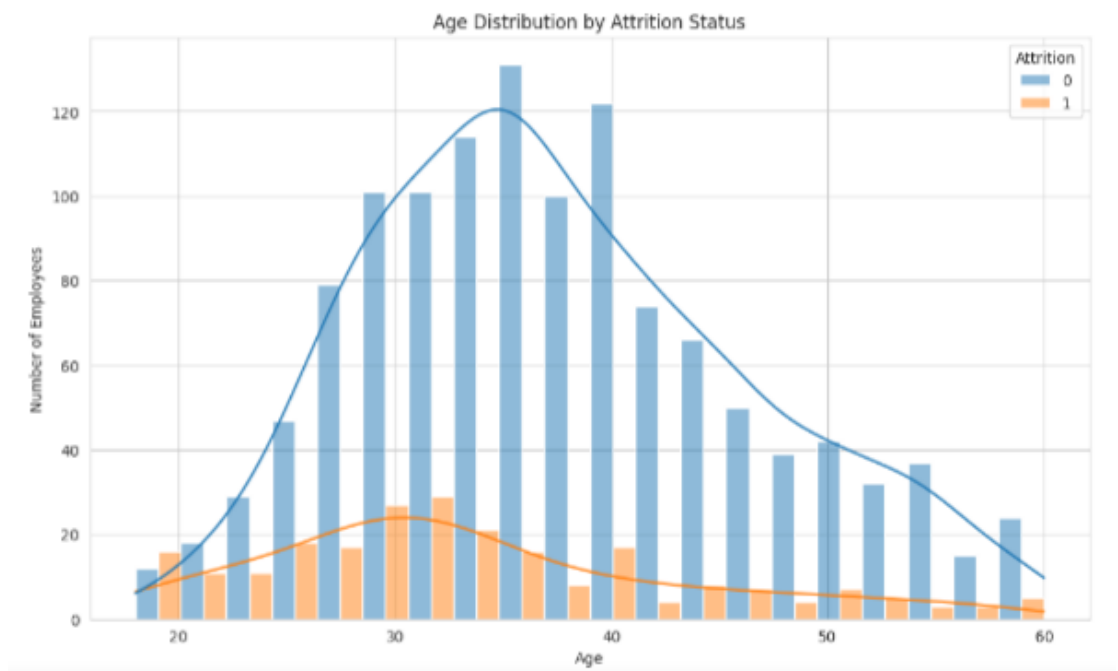
- Attrition Rate by Monthly Income Quartile



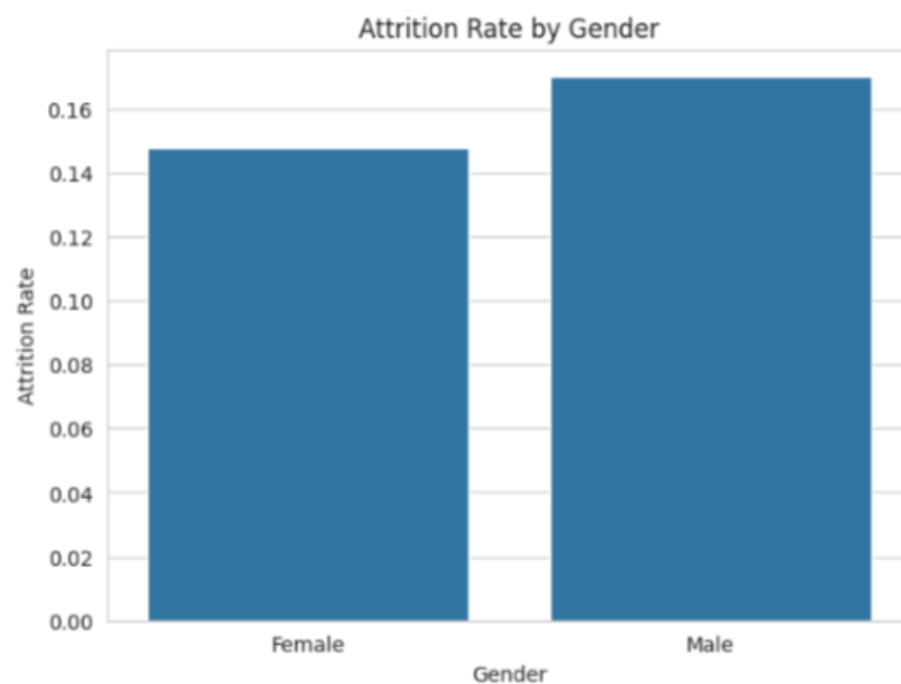
- Attrition Rate by OverTime



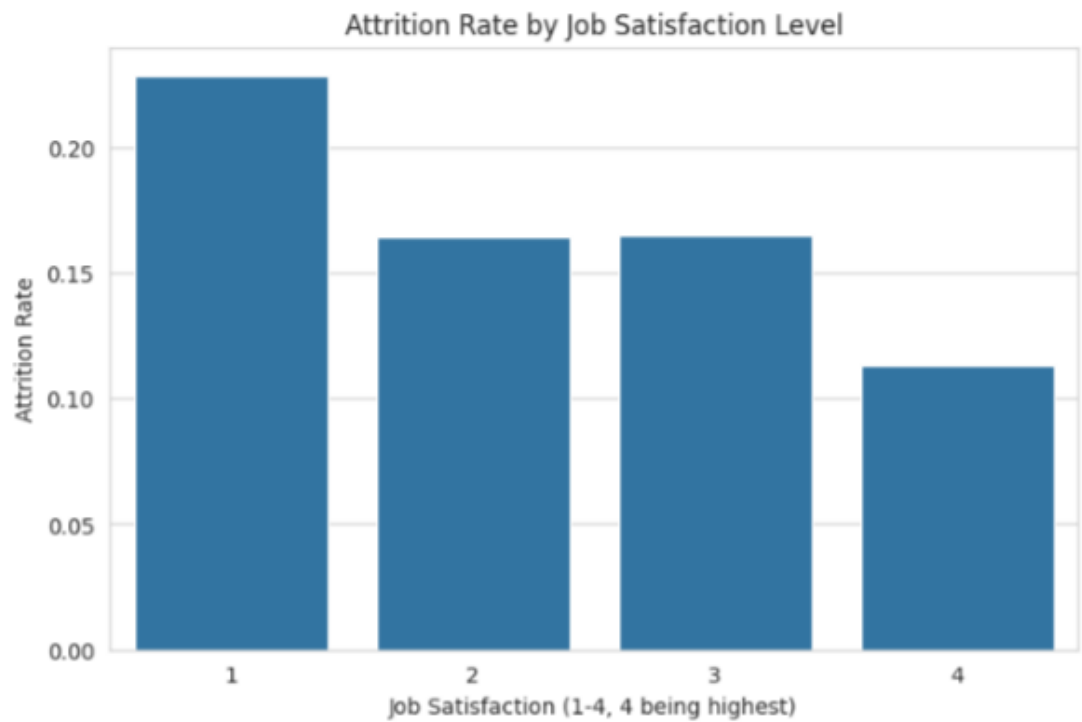
- Age Distribution by Attrition Status



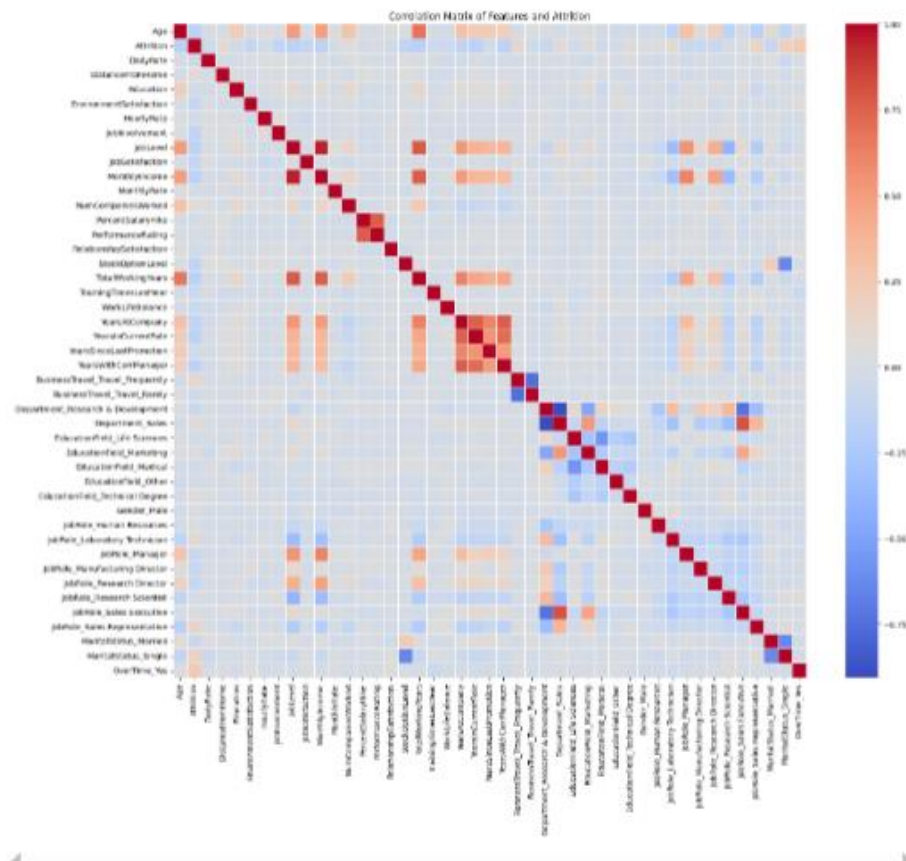
- Attrition Rate by Gender



- Job satisfaction analysis



- Correlation analysis



Important Visualizations:

- Bar Charts
- Heatmaps
- Distribution Plots

The EDA phase helped identify important variables contributing to employee attrition.

9. SQL Business Analysis

Advanced SQL queries were used to perform business-focused HR analysis.

SQL Analysis Included:

- Department-wise attrition rates

Department	Total_Employees	Attrition_Count	Attrition_Rate
Sales	446	92	20.63
Human Resources	63	12	19.05
Research & Development	961	133	13.84

- Average salary of employees leaving

Avg_Salary_Attrited_Employees
-0.36

- Overtime employee analysis

OverTime_Yes	Total_Employees	Attrition_Count	Attrition_Rate
True	416	127	30.53
False	1054	110	10.44

- Age-group risk analysis

Age_Group	Total_Employees	Attrition_Count	Attrition_Rate
Under 25	97	38	39.18
25-34	554	112	20.22
55+	69	11	15.94
45-54	245	25	10.20
35-44	505	51	10.10

- Attrition trends by Years at Company

YearsAtCompany	Total_Employees	Attrition_Count
0	44	16
1	171	59
2	127	27
3	128	20
4	110	19
5	196	21

Advanced SQL Concepts Used:

- Common Table Expressions (CTEs)

Risk_Level	Total_Employees	Attrition_Count
Low Risk	1470	237

- Window Functions

Age	Attrition	Cumulative_Attrition
18	1	4
18	0	4
18	1	4
18	0	4
18	1	4
18	0	4

- Ranking Functions

EmployeeNumber	MonthlyIncome	OverTime_Yes	Attrition	Risk_Rank
701	-1.1673431137498147	True	1	1
1876	-1.1520446508250117	False	1	2
1928	-1.149919864307678	False	1	3
243	-1.1475825991386108	False	1	4
1273	-1.1441829407108768	True	1	5
1433	-1.113798493513004	False	1	6
1427	-1.0929755856431334	False	1	7
1692	-1.0857513114841986	False	1	8
1079	-1.080864302494331	False	1	9
405	-1.0800143878873973	False	1	10

- Views
- Stored Procedures

Department	Total_Employees	Attrition_Count	Attrition_Rate
Sales	446	92	20.63
Research & Development	961	133	13.84
Human Resources	63	12	19.05

SQL analysis provided deeper business insights and improved dashboard reporting capabilities.

10. Dashboard Analysis

An interactive HR Attrition Dashboard was developed using Microsoft Power BI to visualize key workforce metrics.





Dashboard Features:

- KPI Cards
- Interactive Filters
- Department Analysis
- Salary Analysis
- Attrition Trends
- Overtime Insights
- Employee Demographics

Dashboard KPIs:

- Total Employees
- Attrition Count
- Attrition Rate
- Average Salary
- Average Employee Age
- Average Tenure

The dashboard enables HR managers to interactively explore workforce data and identify attrition patterns.

11. Key Insights

The analysis generated several meaningful business insights:

- Employees working overtime showed higher attrition rates
- Lower salary employees were more likely to leave
- Sales and Research departments experienced higher turnover
- Employees with shorter tenure had greater attrition risk
- Younger employees showed higher turnover compared to senior employees
- Work-life balance strongly influenced employee retention

These findings help organizations identify high-risk workforce segments.

12. Business Recommendations

Based on the analysis, the following recommendations are suggested:

Employee Retention Strategies

- Reduce excessive overtime
- Improve employee engagement programs
- Increase career development opportunities
- Review compensation structures
- Improve work-life balance policies

HR Management Improvements

- Monitor high-risk departments regularly
- Conduct employee satisfaction surveys
- Develop retention-focused HR strategies
- Implement employee wellness initiatives

These strategies can help reduce turnover and improve workforce stability.

13. Conclusion

This project successfully analyzed employee attrition patterns using data analytics and business intelligence techniques. The integration of Python, SQL, Excel, and Power BI enabled the creation of a complete end-to-end HR analytics solution.

The findings demonstrated that employee attrition is influenced by multiple factors including salary, overtime, job satisfaction, and tenure. The developed dashboard provides organizations with a powerful tool for monitoring employee trends and making informed HR decisions.

This project also demonstrates practical industry-level skills in:

- Data Cleaning
- Business Analysis
- SQL Analytics
- Dashboard Development
- Data Visualization
- Business Reporting

14. Future Improvements

Future enhancements for this project may include:

- Machine Learning-based attrition prediction
- Real-time HR analytics integration
- Automated reporting systems

- Advanced employee segmentation
- AI-powered HR insights
- Cloud database integration
- Predictive workforce analytics

These improvements can further strengthen workforce decision-making capabilities.